

# AKSHITA SURE

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## EXPERIENCE

### Systems Intern

**D.E. Shaw India Private Limited**

📅 May 2026 — July 2026

- Designed and developed real-time LLM-powered transcription through a headless bot architecture for Webex meetings.
- Engineered low-latency audio ingestion pipelines using WebRTC and the Web Audio API to stream meeting audio for real-time speech recognition.
- Leveraged Redis for transient state management and deployed scalable Kubernetes infrastructure supporting 100+ Webex concurrent meetings.

### Google Summer of Code 2025 Contributor

**GSoC, NumFOCUS, NetworkX**

📅 May 2025 – August 2025

Pull Requests: **NetworkX** | **Nx-parallel**

- Enhanced **nx-parallel**, a parallel backend for **NetworkX** via **joblib**.
- Parallelized **19+** embarrassingly parallel algorithms, achieving **5x speedups** while preserving backward compatibility.
- Improved performance reliability by introducing adaptive execution, strengthening ASV benchmarks and timing infrastructure.
- **Official Maintainer, nx-parallel (From Jan 2026)**

## PROJECTS

### Finora

**Python, FastAPI, TypeScript, Next.js, LangChain**

📅 March 2026

- Engineered a real-time virtual trading simulator with stochastic stock price modeling and dynamic market volatility.
- Built a full-stack trading system with trade execution, portfolio tracking, and a Groq-powered AI mentor.

### Kiosk

**Python, FastAPI, TinyLlama, Whisper, Piper, pgvector**

📅 November 2025

- Developed a privacy-first AI kiosk running fully offline LLM inference using quantised TinyLlama 1.1B for college.
- Implemented a RAG pipeline with IVFFlat indexing and modular pipelines for domain-specific knowledge bases.

### Previously On Netflix

**Flask, Python, JavaScript**

📅 December 2024

- Built a website to analyze Netflix viewing history and deliver personalized insights using TF-IDF-based feature extraction for content recommendations with a 95 % accuracy.
- Integrated engaging features like Netflix Blend for friend-based suggestions and Roast Bot, a chatbot that humorously critiques user binge patterns.

## EDUCATION

Bachelor of Technology in Computer Science and Engineering

**Indian Institute of Information Technology, Gwalior**

📅 2023 - 2027

CGPA: 8.98

## TECHNICAL SKILLS

- **Languages**  
C, C++, Python
- **Frameworks and Tools**  
Flask, Node.js, FastAPI, Github, Keras, Jupyter, Docker, Kubernetes
- **Databases**  
MySQL, PostgreSQL, Microsoft SQL Server
- **Proficiencies**  
Data Structures & Algorithms, Object-Oriented Programming, Machine Learning, Deep Learning, Database Management System, Operating Systems

## ACHIEVEMENTS

- **EuroSciPy 2026** speaker, presenting a technical talk on optimizing parallel graph algorithms
- **DSCI Cyber Security Innovation Challenge**, Finalist
- **Amazon ML Summer School** scholar, top 2% from 1,50,000+ participants
- **JEE Main 2023** – Secured 98.55 percentile nationwide (NTA)

## MENTORSHIPS

Google Women Engineers Scholar

**TalentSprint, supported by Google**

📅 February 2024

- Selected for Cohort 6 of the Women Engineers Program by TalentSprint, supported by Google as one of the top 1% applicants from over 30,000+ applications.

## EXTRA-CURRICULAR

- Trained in Carnatic music for over **10** years; performing at multiple inter-college and cultural events.
- Led the logistics team for ByteForge – a **competitive programming contest** at Infotsav, attracting 200+ participants from across India.
- Authored the blogs: **Women Engineers Program**, **Cracking GSoC** as a 2nd year Btech student.